NephroTwin

A personal digital twin of kidney function for early detection and management of chronic kidney disease

[Team name] · [Institute]

CKD is silent, common in India, and judged one noisy result at a time

101 M

Indians with diabetes (ICMR-INDIAB-17, 2023)

136 M

with prediabetes

~315 M

with hypertension, the other leading cause of CKD

- Kidney function falls for years without symptoms; many people are diagnosed late.
- A single eGFR value is noisy: a real decline is easy to miss and a one-off bad result is easy to over-react to.
- Clinicians need the trajectory: where this person is heading, how sure we are, and what would change it.

A twin, not a risk score: six properties shown in the demo

Personal

Own state per patient: current eGFR and slope.

Continuously updated

Bayesian (Kalman) update after every result.

Predictive

Forecast eGFR; P(eGFR < 30) and P(eGFR < 15).

Simulatable

What-if care plans with trial-based effects.

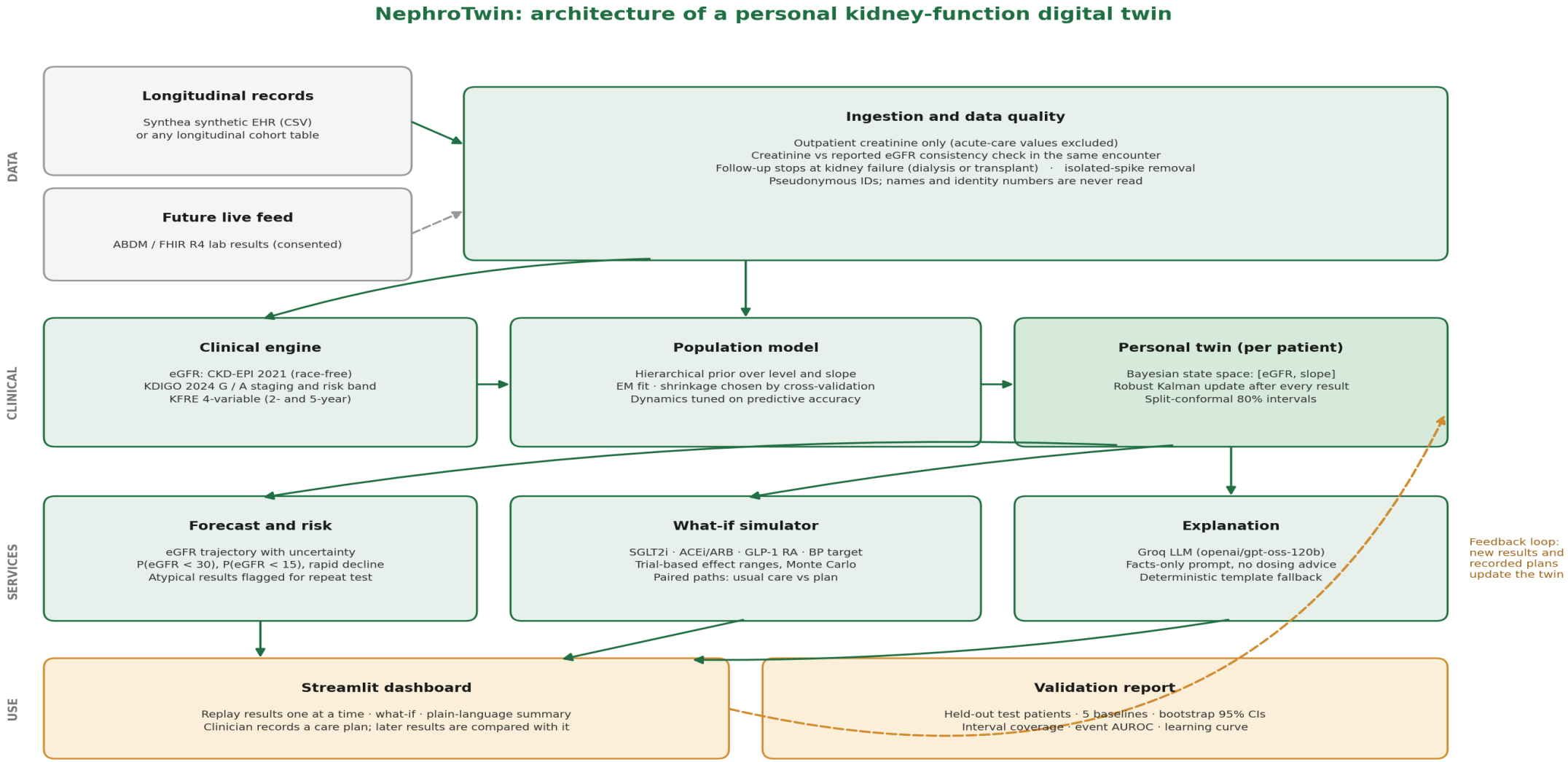
Honest

Conformal 80% intervals, checked on unseen patients.

Two-way

Recorded plans are compared with later results.

Architecture



How the twin works

Clinical engine	eGFR by CKD-EPI 2021 (race-free) · KDIGO 2024 staging · KFRE 4-variable where valid
Population model	Hierarchical prior over starting eGFR and slope · EM fit · shrinkage chosen by cross-validation
Personal twin	State-space model [eGFR, slope] · robust Kalman update · atypical results flagged for repeat test
What-if simulator	SGLT2i, ACEi/ARB, GLP-1 RA, BP target · triangular effect ranges from trials · paired Monte Carlo
Explanation	Groq LLM (openai/gpt-oss-120b) writes patient and clinician notes from computed facts only · template fallback

Data: synthetic patients, cleaned like real records

404

patients after cleaning (Synthea v4.0.0, ages 40-85)

4,274

outpatient creatinine results

9 yrs

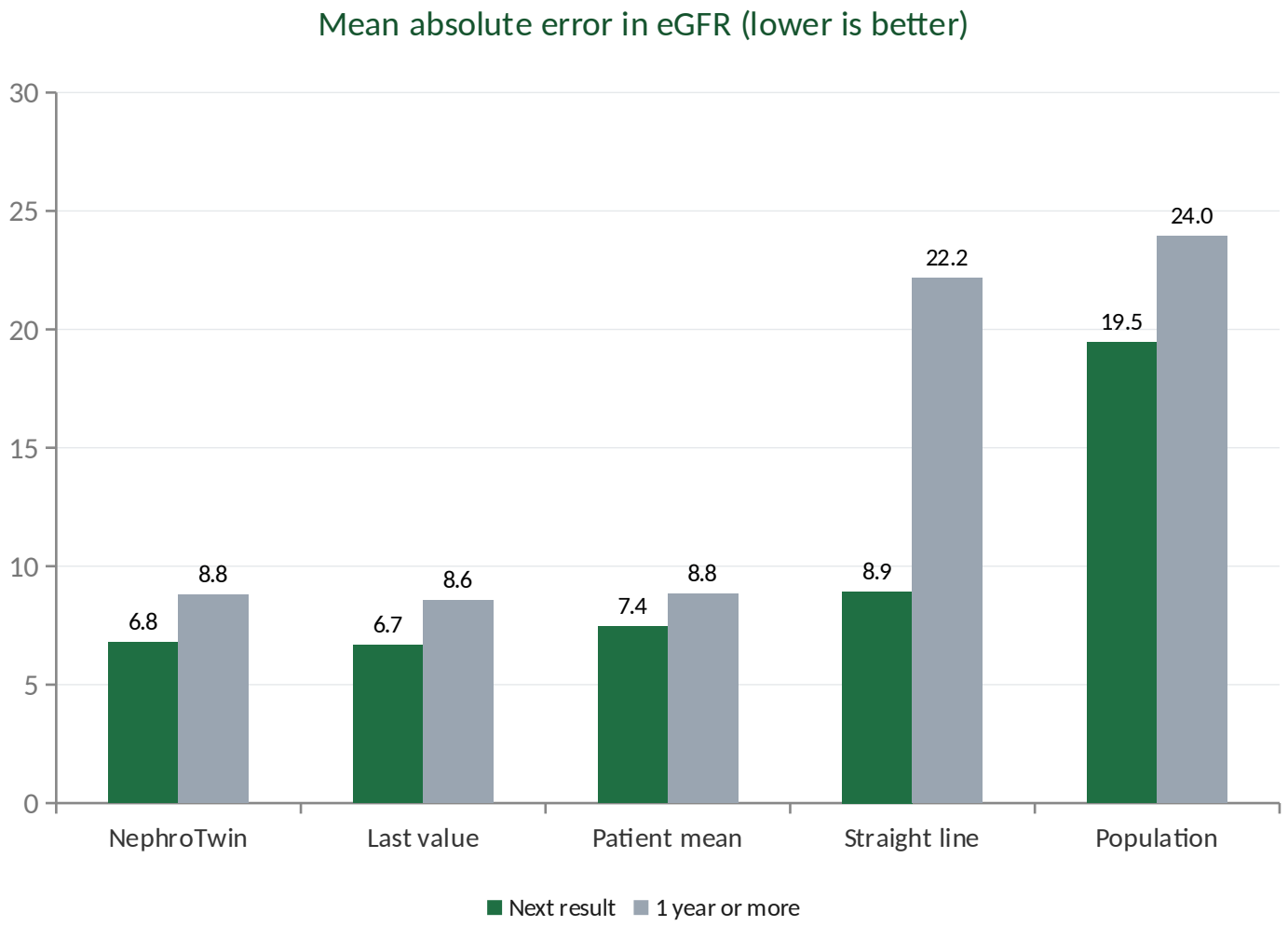
median follow-up per patient

21%

CKD G3 or worse at entry

- Acute-care results excluded (6,712): hospital values reflect acute kidney injury, not chronic function.
- Creatinine that contradicts the lab eGFR in the same visit removed (7,412). Without this, healthy people look like severe CKD.
- Follow-up stops at dialysis or transplant (1,541 results); isolated spikes removed (103).
- Names and identity numbers are never read; only pseudonymous IDs leave the pipeline.

Results on 121 unseen test patients



Beats straight-line extrapolation

2.1 points better next result; 13.4 points better at 1 year+ (95% CIs exclude zero).

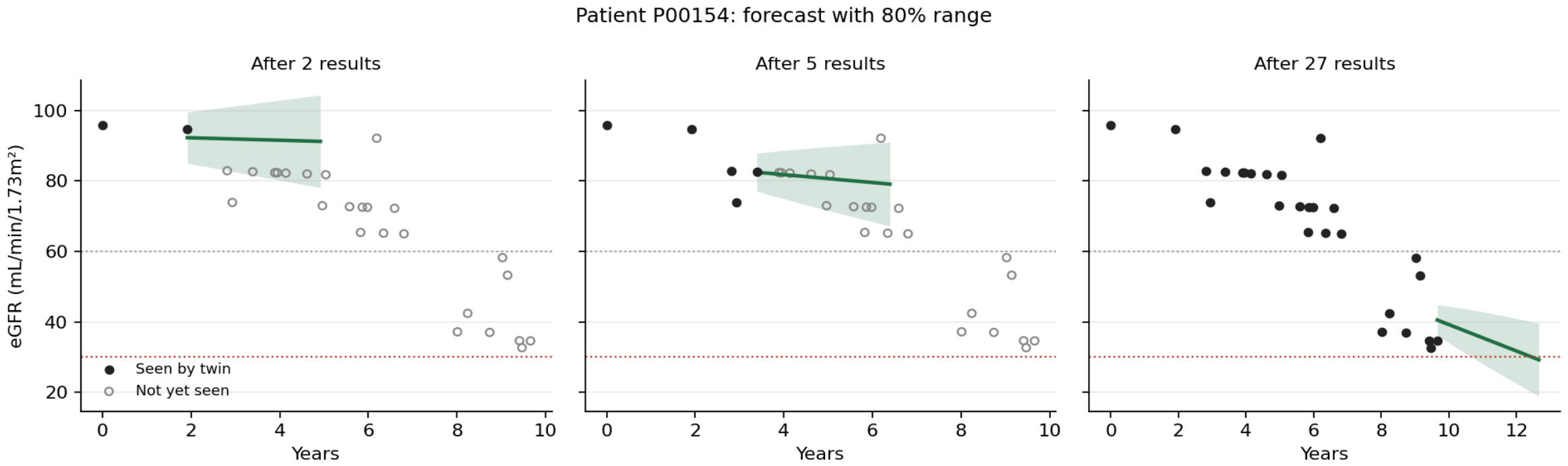
Ties last value carried forward

Difference not significant. Stable synthetic cohort makes this baseline strong.

Honest uncertainty

82% of next results and 84% at 1 year+ fell inside the 80% interval.

The twin learns each patient as results arrive



- After 2 results the forecast leans on the population and is wide; after 5 it narrows; after 27 it tracks the decline and projects crossing eGFR 30.
- Event prediction (eGFR below 30 at about 2 years): AUROC 0.93 vs 0.58 for the population model, but only 5 events, so preliminary.

What-if care plans and plain-language summaries

Scenario	Slower decline (range)	Evidence
SGLT2 inhibitor	30% to 55%	DAPA-CKD, EMPA-KIDNEY, DAPA advKD
ACE inhibitor / ARB optimisation	5% to 25%	RENAAL, IDNT
GLP-1 RA (semaglutide)	20% to 40%	FLOW
Blood pressure to target	0% to 20%	KDIGO 2021 BP guideline

Scenarios show trial averages applied to this twin, not personal predictions of treatment response.

Usual care vs plan

Both arms use identical random draws, so the gap reflects the plan. Output: eGFR difference and kidney-failure probability, each with a range.

Explanation

Patient and clinician summaries from computed numbers only. No diagnosis, no dosing. Works without a key through a fixed template.

Limitations, safeguards and next steps

Limitations

Synthetic US-modelled data with mostly stable kidney function.

Few progression events, so event metrics are preliminary.

Treatment effects are trial averages.

Not yet validated on Indian patients.

Safeguards

No real patient data; identity fields never read.

Atypical results flagged for repeat testing.

KFRE labelled outside its validated range.

Decision support only; shown on every screen.

Next steps

Validate on an Indian CKD cohort with ethics approval.

Calibrate eGFR and KFRE for Indian populations.

Connect to ABDM-linked lab results through FHIR, under DPDP Act 2023 consent rules.